

Nyasha Hama

Backend & Platform Engineer | Go, TypeScript, Rust | Healthcare Ops, Data Systems & Upstream OSS

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Summary

Backend and platform engineer building public, working systems in Go, TypeScript, Rust, PostgreSQL, and Redis, with emphasis on healthcare operations, property platforms, data workflows, auth/RBAC, observability, and release gates.

Recent evidence includes merged upstream Turso database-engine fixes, a merged CrossHair Python/C tracer contribution, a live clinic-operations platform for South Africa's public healthcare network, and a live sectional-title property platform with real workflow depth.

Skills

Languages: Go, TypeScript, Rust, SQL, JavaScript/Node.js, Java, C++

Backend, Data & Platform: REST APIs, service design, JWT auth, RBAC, PostgreSQL, Redis, pgx, sqlc, migrations, indexing, SQLite internals

Infrastructure & Quality: Docker, Linux, Prometheus, structured logging, request tracing, GitHub Actions, CI/CD, E2E tests, smoke/load checks, release gates

Experience

Backend & Platform Engineer, Independent Software Engineer – Remote Jan 2023 – present

- Built and shipped public systems across healthcare operations, property management, property intelligence, policy enforcement, and backend tooling.
- Designed Go APIs, TypeScript product surfaces, Rust infrastructure/database patches, PostgreSQL schemas, Redis workflows, Dockerized environments, CI checks, observability, and release-readiness gates.

Open Source Contributions

Upstream Contributor, Turso May 2026

- Merged two Rust database-engine fixes around SQLite-compatible schema rewrites, preserving AUTOINCREMENT after DROP COLUMN and clearing stale sqlite_sequence metadata after ALTER COLUMN removes an AUTOINCREMENT rowid alias.
- Patched schema serialization, ALTER execution state, SQL regressions, and file-backed reopen integration coverage for rowid reuse and stale metadata failures.
- Links: [PR #6993](#) | [PR #7117](#)

Upstream Contributor, CrossHair May 2026

- Merged Python/C tracer work moving call target normalization into CrossHair's C extension while keeping keyword handling and trace dispatch behavior stable.
- Added coverage for bound Python methods, callable instances, C-level bound methods, reference behavior, and descriptor-error propagation.
- Links: [PR #413](#)

Projects

ClinicPulse

Live TypeScript and Go clinic-operations platform for South Africa's public primary healthcare network, covering facility status, referral routing, field reporting, and district operational intelligence.

- Built role-based workflows for admin, district, facility, field, partner, and public users, with PostgreSQL-backed API paths and product surfaces for operational decision-making.
- Added release gates, auth/security hardening, pilot data integrity, partner workflows, E2E coverage, smoke/load scripts, and observability primitives including request correlation, metrics, and runbooks.
- Links: [GitHub](#) | [Live](#)

StrataHQ

Live Next.js and Go platform for South African sectional-title property operations, serving agents, trustees, and residents across scheme management workflows.

- Built scheme-scoped levy operations, maintenance workflows, resident/trustee communications, AGM administration, document access, audit logs, open API access, and predictive levy analytics.
- Modeled production-style backend paths with PostgreSQL, pgx, sqlc, goose migrations, Redis, auth/RBAC, tests/CI, and deployment-ready project structure.
- Links: [GitHub](#) | [Live](#)

Healthcare Access Connector

Go healthcare access backend connecting patients and providers through scheduling, search, notifications, telemedicine support, and role-aware workflows.

- Designed patient, provider, appointment, staff invitation, and notification flows with PostgreSQL-backed domain modeling, validation, and explicit service boundaries.
- Implemented JWT/RBAC, rate limiting, Redis caching, NATS messaging, Docker local infrastructure, Prometheus metrics, structured logging, health endpoints, and graceful shutdown.
- Links: [GitHub](#)

TitleChain

Property transaction intelligence platform for South African conveyancing, banking, and insurance workflows.

- Built a monorepo foundation spanning a Next.js portal, Go API/control plane, Rust data-processing worker, PostgreSQL migrations, and Docker infrastructure.
- Added case workflow, property projection, source ingestion, decision engine, and pilot-customer verification paths for historically versioned property intelligence.
- Links: [GitHub](#) | [Live](#)

Education

University of Johannesburg — BSc Computer Science Coursework, completed semesters through final-year coursework, 2022–2024